Problem 3

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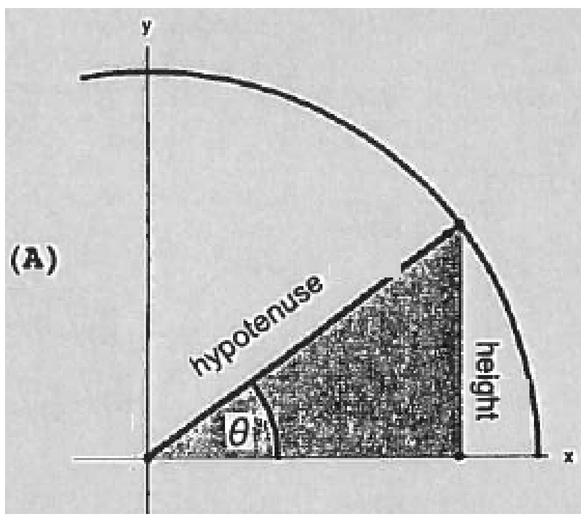
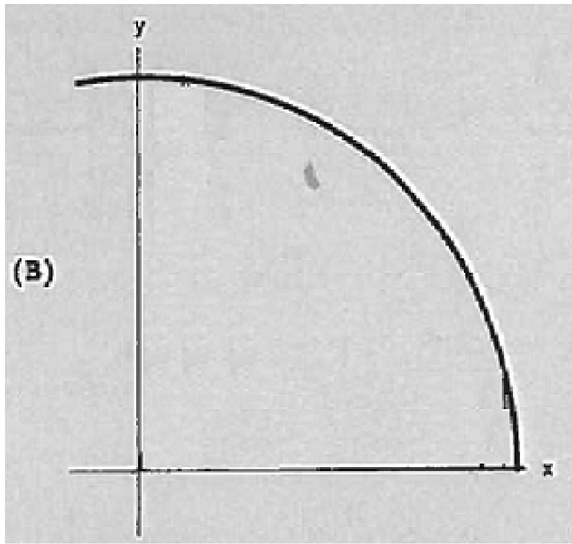
Problem 1 *A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see Figure (A)). One of its sides lies on the x-axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x-axis. Assume that the angle θ changes at the rate $\frac{d\theta}{dt} = 0.2$ radians per second.*



(a) *Label Figure A.*

(b) *In Figure B, draw the the triangle twice; once when θ is small and once more, when θ is close to $\frac{\pi}{2}$.*



(c) *Find the rate of change of the height of the triangle when* $\theta = \dfrac{\pi}{3}$

(d) *Find the rate of change of the area of the triangle when* $\theta = \dfrac{\pi}{3}$